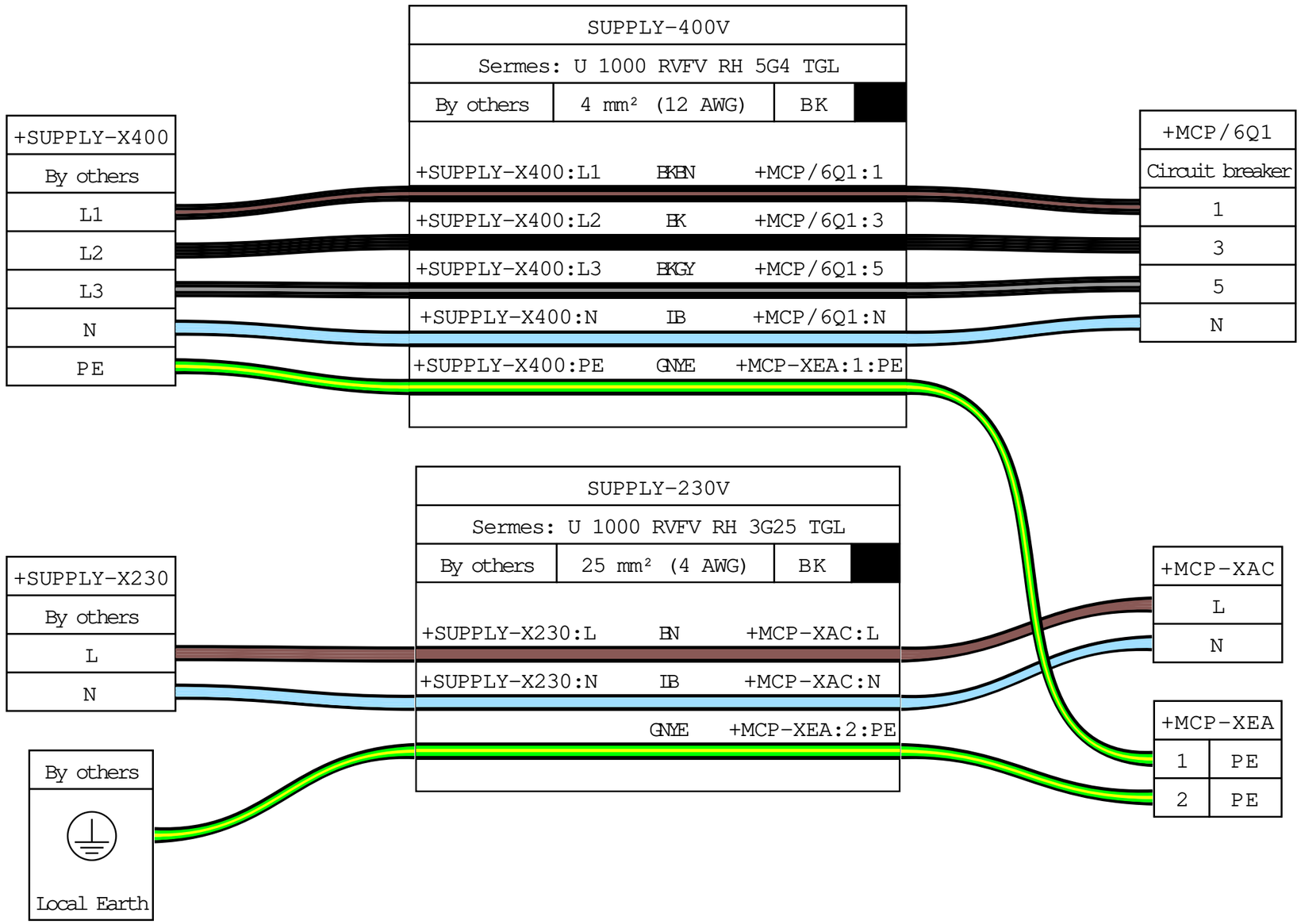
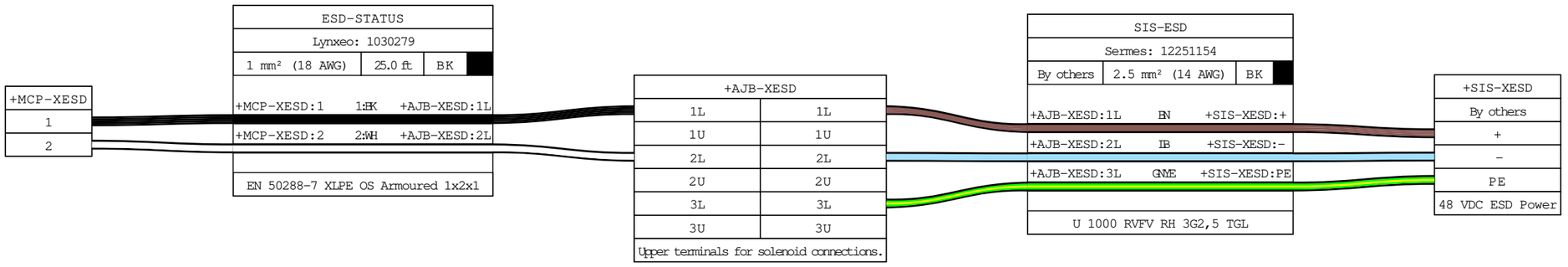

SUPPLY-400V
Sermes: U 1000 RVFV RH 5G4 TGL
By others
4 mm² (12 AWG)
BK
+SUPPLY-X400
By others
L1
L2
L3
N
PE
+SUPPLY-X400:L1
BKBN
+MCP/6Q1:1
+SUPPLY-X400:L2
BK
+MCP/6Q1:3
+SUPPLY-X400:L3
BKGY
+MCP/6Q1:5
+SUPPLY-X400:N
LB
+MCP/6Q1:N
+SUPPLY-X400:PE
GNYE
+MCP-XEA:1:PE
+MCP/6Q1
Circuit breaker
1
3
5
N
SUPPLY-230V
Sermes: U 1000 RVFV RH 3G25 TGL
By others
25 mm² (4 AWG)
BK
+SUPPLY-X230
By others
L
N
+SUPPLY-X230:L
BN
+MCP-XAC:L
+SUPPLY-X230:N
LB
+MCP-XAC:N
GNYE
+MCP-XEA:2:PE
+MCP-XAC
L
N
+MCP-XEA
1
PE
2
PE
By others
Local Earth


+MCP-XESD
1
2
ESD-STATUS
Lynxeo: 1030279
1 mm² (18 AWG)
25.0 ft
BK
+MCP-XESD:1
1:BK
+AJB-XESD:1L
+MCP-XESD:2
2:WH
+AJB-XESD:2L
EN 50288-7 XLPE OS Armoured 1x2x1
+AJB-XESD
1L
1L
1U
1U
2L
2L
2U
2U
3L
3L
3U
3U
Upper terminals for solenoid connections.
SIS-ESD
Sermes: 12251154
By others
2.5 mm² (14 AWG)
BK
+AJB-XESD:1L
BN
+SIS-XESD:+
+AJB-XESD:2L
IB
+SIS-XESD:-
+AJB-XESD:3L
GNYE
+SIS-XESD:PE
U 1000 RVFV RH 3G2,5 TGL
+SIS-XESD
By others
+
-
PE
48 VDC ESD Power

| +DCS-INSTRUMENT | |
|---|---|
| By others | |
| ZC + | 1 |
| ZC – | 2 |
| ZI + | 3 |
| ZI – | 4 |
| PT ACC + | 5 |
| PT ACC – | 6 |
| PT A + | 7 |
| PT A – | 8 |
| PT B + | 9 |
| PT B – | 10 |
| ESD Alarm + | 11 |
| ESD Alarm – | 12 |
| Auto Alarm + | 13 |
| Auto Alarm – | 14 |
| Trouble Alarm + | 15 |
| Trouble Alarm – | 16 |
| Mode Bit 0 + | 17 |
| Mode Bit 0 – | 18 |
| Mode Bit 1 + | 19 |
| Mode Bit 1 – | 20 |
| Mode Bit 2 + | 21 |
| Mode Bit 2 – | 22 |
| SHIELD | 23 |

Shield grounded at DCS end only
Unused cores terminated and earthed at DCS end per 60079-14 8.2.3.2

| DCS-INSTRUMENT | | | |
|---|---|---|---|
| Sermes: 14483064 | | | |
| By others | 0.88 mm² (Unknown AWG) | + S | BU |

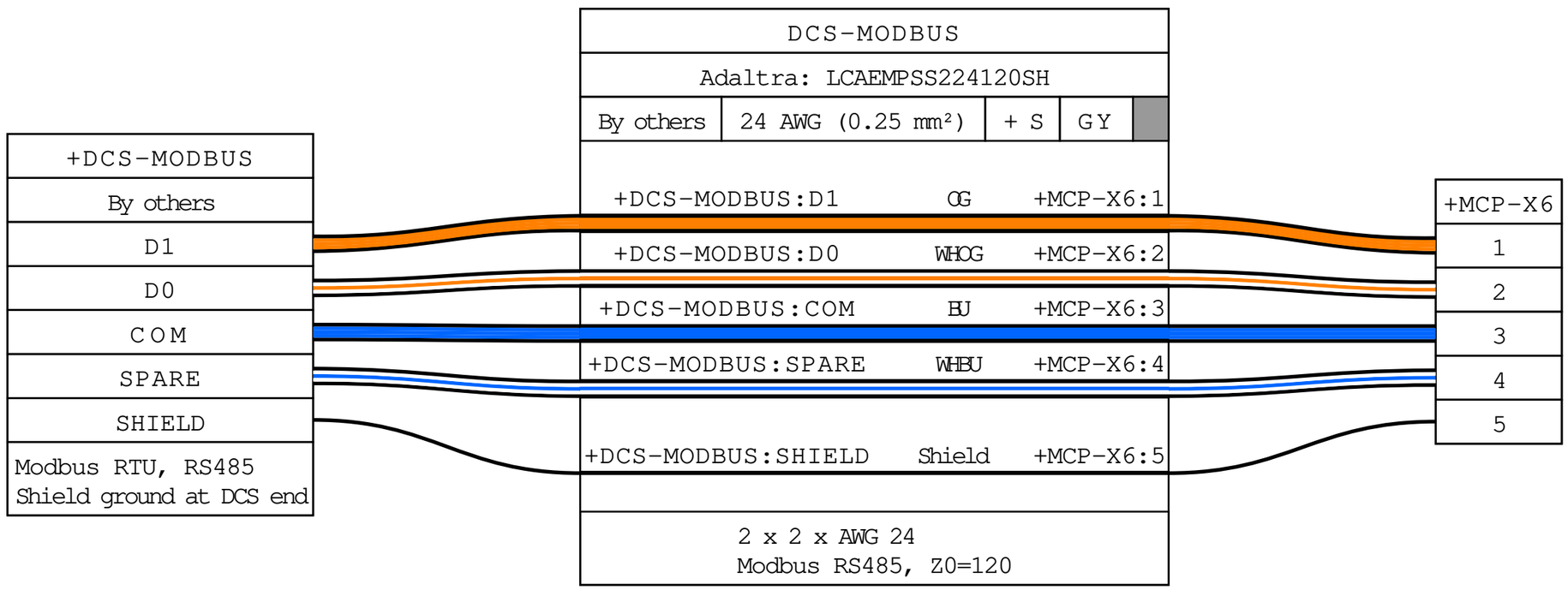
| | | |
|---|---|---|
| +DCS-INSTRUMENT:1:ZC + | RD:PR1 | +MCP-X5:1U |
| +DCS-INSTRUMENT:2:ZC – | WH:PR1 | +MCP-X5:1M |
| +DCS-INSTRUMENT:3:ZI + | RD:PR2 | +MCP-X5:2U |
| +DCS-INSTRUMENT:4:ZI – | WH:PR2 | +MCP-X5:2M |
| +DCS-INSTRUMENT:5:PT ACC + | RD:PR3 | +MCP-X5:3U |
| +DCS-INSTRUMENT:6:PT ACC – | WH:PR3 | +MCP-X5:3M |
| +DCS-INSTRUMENT:7:PT A + | RD:PR4 | +MCP-X5:4U |
| +DCS-INSTRUMENT:8:PT A – | WH:PR4 | +MCP-X5:4M |
| +DCS-INSTRUMENT:9:PT B + | RD:PR5 | +MCP-X5:5U |
| +DCS-INSTRUMENT:10:PT B – | WH:PR5 | +MCP-X5:5M |
| +DCS-INSTRUMENT:11:ESD Alarm + | RD:PR6 | +MCP-X3:1 |
| +DCS-INSTRUMENT:12:ESD Alarm – | WH:PR6 | +MCP-X3:2 |
| +DCS-INSTRUMENT:13:Auto Alarm + | RD:PR7 | +MCP-X3:3 |
| +DCS-INSTRUMENT:14:Auto Alarm – | WH:PR7 | +MCP-X3:4 |
| +DCS-INSTRUMENT:15:Trouble Alarm + | RD:PR8 | +MCP-X3:5 |
| +DCS-INSTRUMENT:16:Trouble Alarm – | WH:PR8 | +MCP-X3:6 |
| +DCS-INSTRUMENT:17:Mode Bit 0 + | RD:PR9 | +MCP-X3:7 |
| +DCS-INSTRUMENT:18:Mode Bit 0 – | WH:PR9 | +MCP-X3:8 |
| +DCS-INSTRUMENT:19:Mode Bit 1 + | RD:PR10 | +MCP-X3:9 |
| +DCS-INSTRUMENT:20:Mode Bit 1 – | WH:PR10 | +MCP-X3:10 |
| +DCS-INSTRUMENT:21:Mode Bit 2 + | RD:PR11 | +MCP-X3:11 |
| +DCS-INSTRUMENT:22:Mode Bit 2 – | WH:PR11 | +MCP-X3:12 |
| +DCS-INSTRUMENT:23:SHIELD | Shield | +MCP-X3:13 |

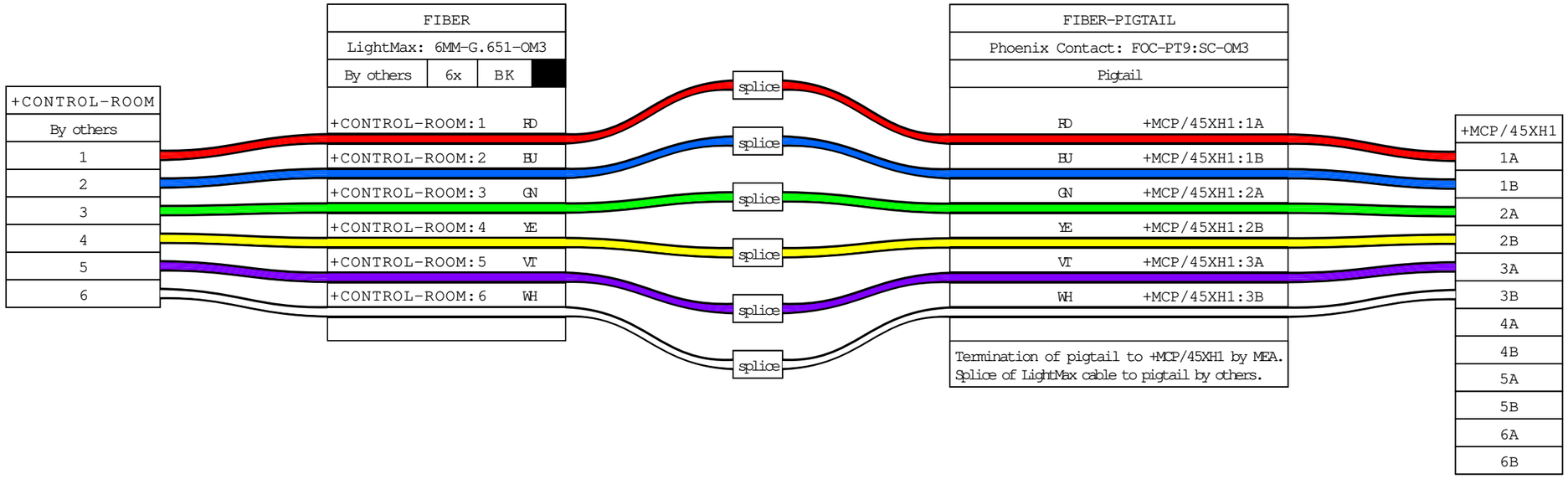
19IP09EGFA – BLEU19IP09EGFA – BLEU
Pairs 12 – 19 unused

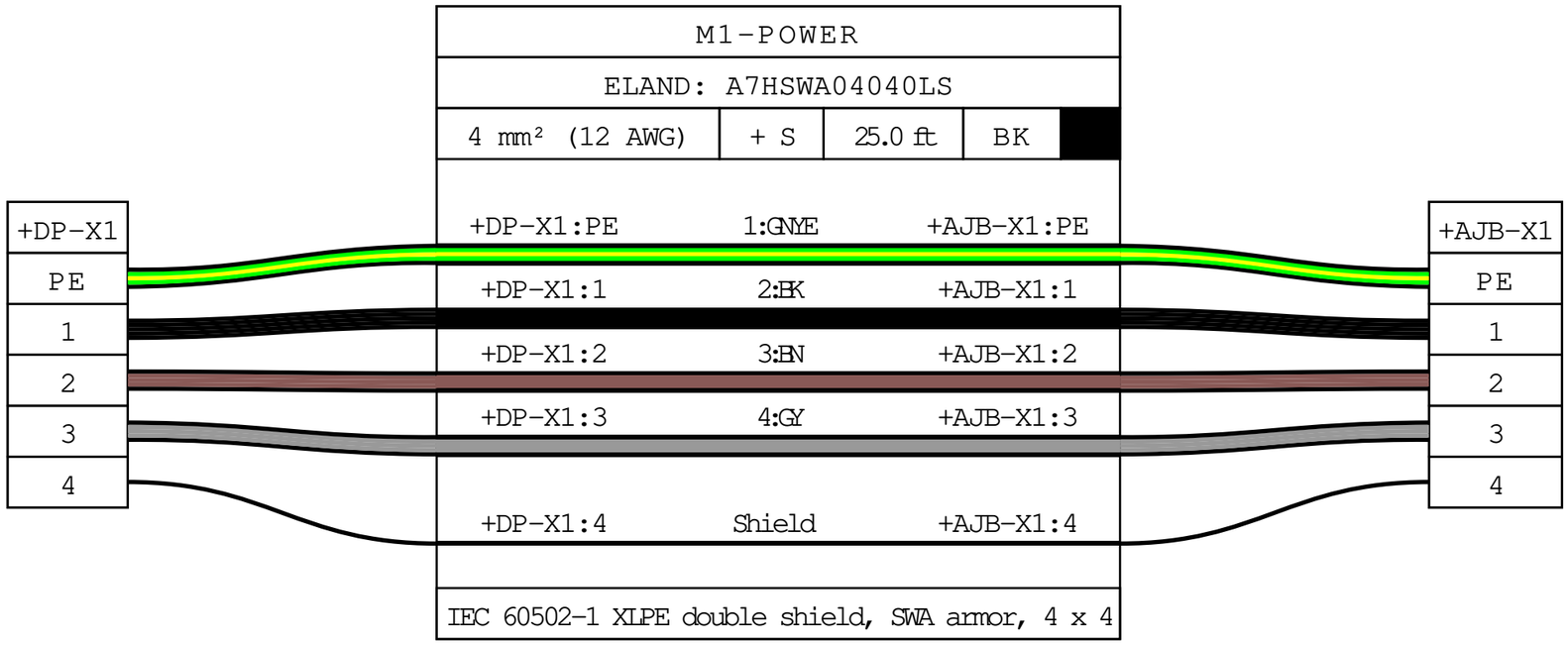
| +MCP-X5 |
|---|
| 1U |
| 1M |
| 2U |
| 2M |
| 3U |
| 3M |
| 4U |
| 4M |
| 5U |
| 5M |

| +MCP-X3 |
|---|
| 1 |
| 2 |
| 3 |
| 4 |
| 5 |
| 6 |
| 7 |
| 8 |
| 9 |
| 10 |
| 11 |
| 12 |
| 13 |

Terminate unused cores per IEC 60079-14 8.2.3.2.


DCS-MODBUS
Adaltra: LCAEMPSS224120SH
By others
24 AWG (0.25 mm²)
+ S
GY
+DCS-MODBUS:D1
OG
+MCP-X6:1
+DCS-MODBUS:D0
WHOG
+MCP-X6:2
+DCS-MODBUS:COM
BU
+MCP-X6:3
+DCS-MODBUS:SPARE
WHBU
+MCP-X6:4
+DCS-MODBUS:SHIELD
Shield
+MCP-X6:5
2 x 2 x AWG 24
Modbus RS485, Z0=120
+DCS-MODBUS
By others
D1
D0
COM
SPARE
SHIELD
Modbus RTU, RS485
Shield ground at DCS end
+MCP-X6
1
2
3
4
5

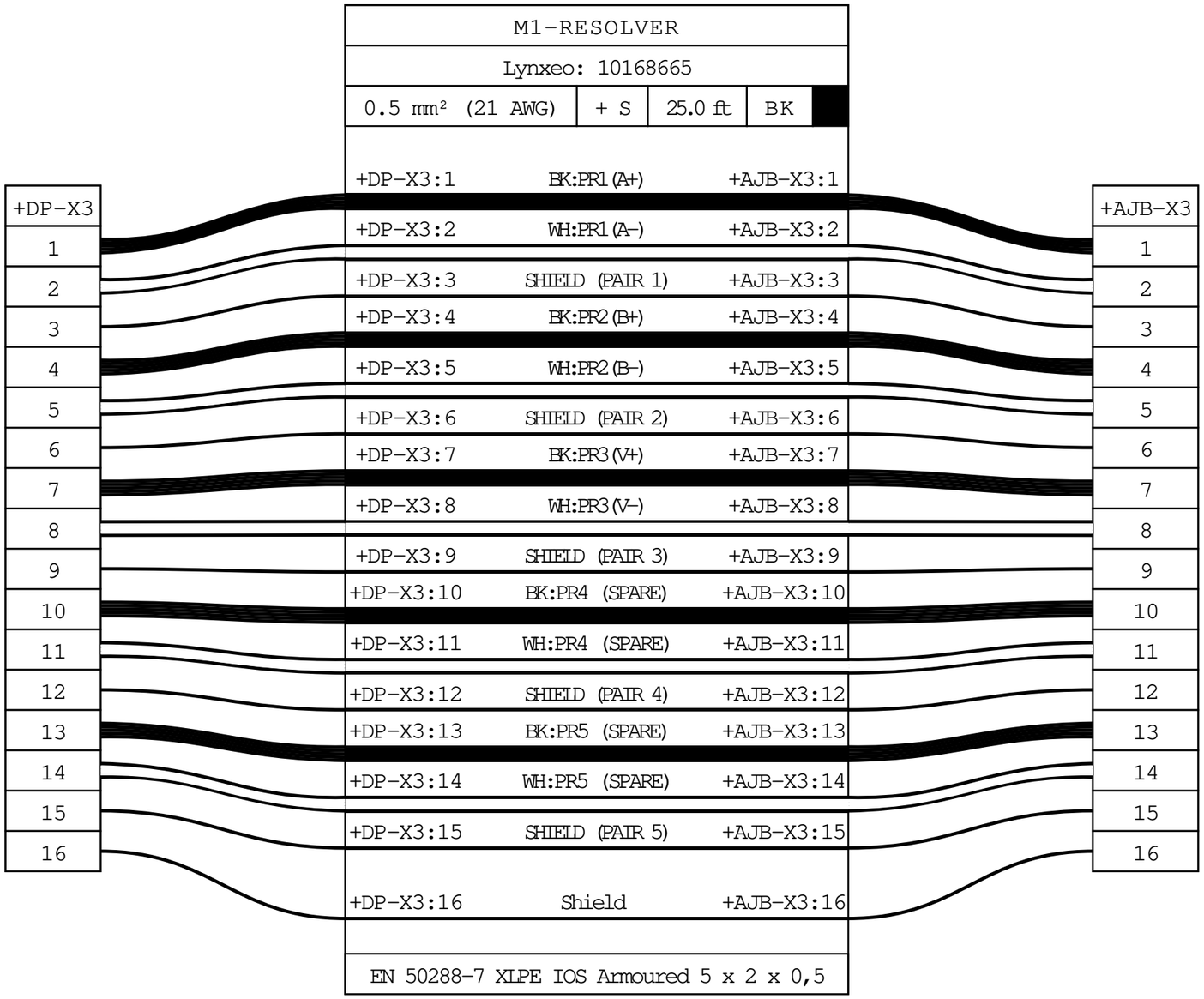

+CONTROL-ROOM
By others
1
2
3
4
5
6
FIBER
LightMax: 6MM-G.651-OM3
By others
6x
BK
+CONTROL-ROOM:1 RD
+CONTROL-ROOM:2 BU
+CONTROL-ROOM:3 GN
+CONTROL-ROOM:4 YE
+CONTROL-ROOM:5 VT
+CONTROL-ROOM:6 WH
splice
splice
splice
splice
splice
splice
FIBER-PIGTAIL
Phoenix Contact: FOC-PT9:SC-OM3
Pigtail
RD +MCP/45XH1:1A
BU +MCP/45XH1:1B
GN +MCP/45XH1:2A
YE +MCP/45XH1:2B
VT +MCP/45XH1:3A
WH +MCP/45XH1:3B
Termination of pigtail to +MCP/45XH1 by MEA.
Splice of LightMax cable to pigtail by others.
+MCP/45XH1
1A
1B
2A
2B
3A
3B
4A
4B
5A
5B
6A
6B

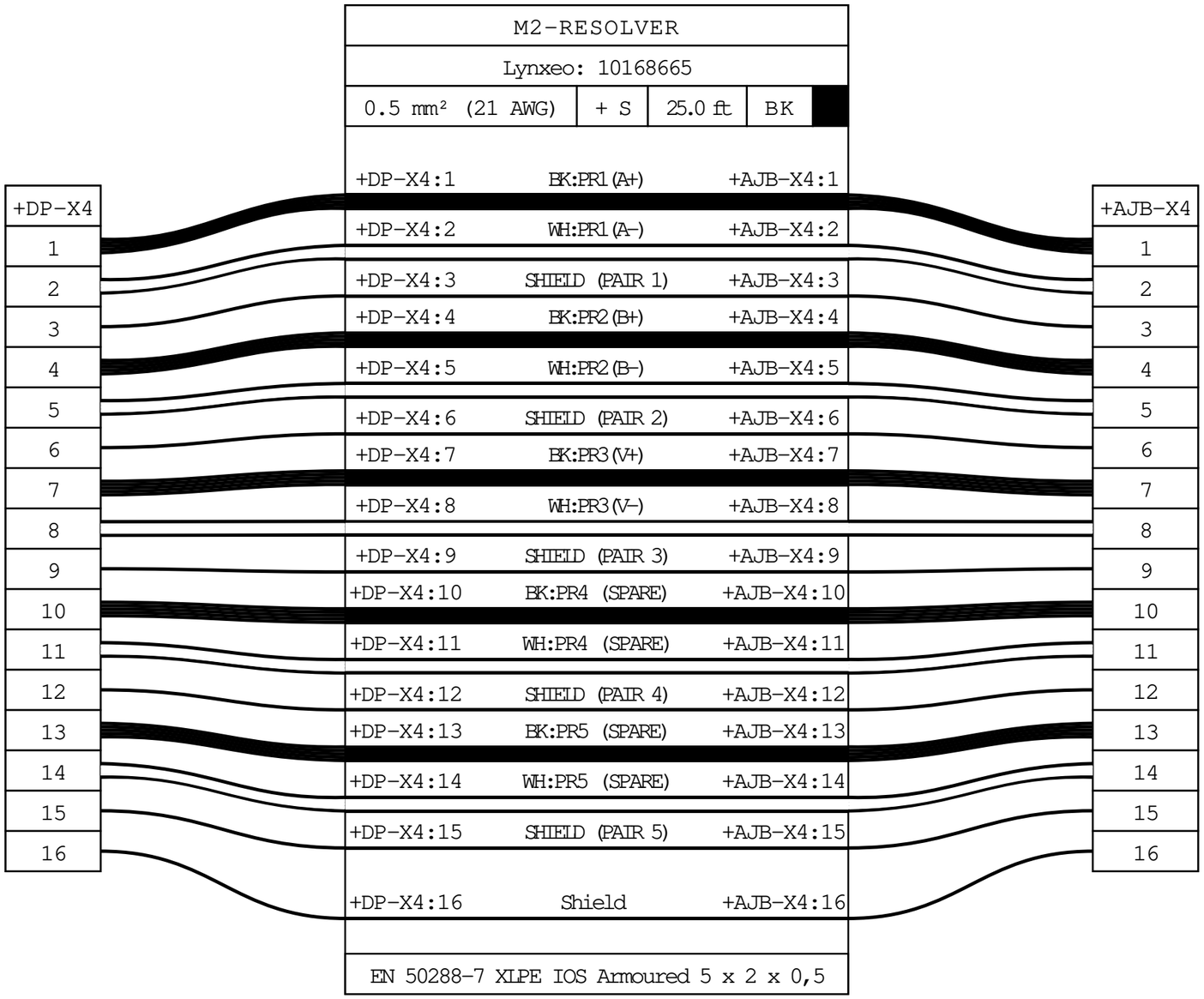

M1-POWER
ELAND: A7HSWA04040LS
4 mm² (12 AWG)
+ S
25.0 ft
BK
+DP-X1
PE
1
2
3
4
+DP-X1:PE
1:GNYE
+AJB-X1:PE
+DP-X1:1
2:BK
+AJB-X1:1
+DP-X1:2
3:BN
+AJB-X1:2
+DP-X1:3
4:GY
+AJB-X1:3
+DP-X1:4
Shield
+AJB-X1:4
IEC 60502-1 XLPE double shield, SWA armor, 4 x 4
+AJB-X1
PE
1
2
3
4

| M2-POWER | | | |
|---|---|---|---|
| ELAND: A7HSWA04040LS | | | |
| 4 mm² (12 AWG) | + S | 25.0 ft | BK |

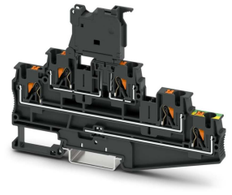
| +DP-X2 | +DP-X2:PE | 1:GNYE | +AJB-X2:PE | +AJB-X2 |
|---|---|---|---|---|
| PE | +DP-X2:1 | 2:BK | +AJB-X2:1 | PE |
| 1 | +DP-X2:2 | 3:BN | +AJB-X2:2 | 1 |
| 2 | +DP-X2:3 | 4:GY | +AJB-X2:3 | 2 |
| 3 | +DP-X2:4 | Shield | +AJB-X2:4 | 3 |
| 4 | | | | 4 |

IEC 60502-1 XLPE double shield, SWA armor, 4 x 4


M1-RESOLVER
Lynxeo: 10168665
0.5 mm² (21 AWG)
+ S
25.0 ft
BK
+DP-X3
+AJB-X3
+DP-X3:1 BK:PR1(A+) +AJB-X3:1
+DP-X3:2 WH:PR1(A-) +AJB-X3:2
+DP-X3:3 SHIELD (PAIR 1) +AJB-X3:3
+DP-X3:4 BK:PR2(B+) +AJB-X3:4
+DP-X3:5 WH:PR2(B-) +AJB-X3:5
+DP-X3:6 SHIELD (PAIR 2) +AJB-X3:6
+DP-X3:7 BK:PR3(V+) +AJB-X3:7
+DP-X3:8 WH:PR3(V-) +AJB-X3:8
+DP-X3:9 SHIELD (PAIR 3) +AJB-X3:9
+DP-X3:10 BK:PR4 (SPARE) +AJB-X3:10
+DP-X3:11 WH:PR4 (SPARE) +AJB-X3:11
+DP-X3:12 SHIELD (PAIR 4) +AJB-X3:12
+DP-X3:13 BK:PR5 (SPARE) +AJB-X3:13
+DP-X3:14 WH:PR5 (SPARE) +AJB-X3:14
+DP-X3:15 SHIELD (PAIR 5) +AJB-X3:15
+DP-X3:16 Shield +AJB-X3:16
EN 50288-7 XLPE IOS Armoured 5 x 2 x 0,5


M2-RESOLVER
Lynxeo: 10168665
0.5 mm² (21 AWG)
+ S
25.0 ft
BK
+DP-X4
+AJB-X4
+DP-X4:1 BK:PR1(A+) +AJB-X4:1
+DP-X4:2 WH:PR1(A-) +AJB-X4:2
+DP-X4:3 SHIELD (PAIR 1) +AJB-X4:3
+DP-X4:4 BK:PR2(B+) +AJB-X4:4
+DP-X4:5 WH:PR2(B-) +AJB-X4:5
+DP-X4:6 SHIELD (PAIR 2) +AJB-X4:6
+DP-X4:7 BK:PR3(V+) +AJB-X4:7
+DP-X4:8 WH:PR3(V-) +AJB-X4:8
+DP-X4:9 SHIELD (PAIR 3) +AJB-X4:9
+DP-X4:10 BK:PR4 (SPARE) +AJB-X4:10
+DP-X4:11 WH:PR4 (SPARE) +AJB-X4:11
+DP-X4:12 SHIELD (PAIR 4) +AJB-X4:12
+DP-X4:13 BK:PR5 (SPARE) +AJB-X4:13
+DP-X4:14 WH:PR5 (SPARE) +AJB-X4:14
+DP-X4:15 SHIELD (PAIR 5) +AJB-X4:15
+DP-X4:16 Shield +AJB-X4:16
EN 50288-7 XLPE IOS Armoured 5 x 2 x 0,5

| +MCP-X4 |
|---|
| 1U |
| 1M |
| 1L |
| 2U |
| 2M |
| 2L |
| 3U |
| 3M |
| 3L |
| 4U |
| 4M |
| 4L |
| 5U |
| 5M |
| 5L |
| SHIELD |



3-deck fuse terminal block with shield to rail.

ACTUATOR-ANALOG

Lynxeo: 10168748

| 0.75 mm² (20 AWG) | + S | 25.0 ft | BK |
|---|---|---|---|

| | | |
|---|---|---|
| +MCP-X4:1U | BK:1(BK) | +AJB-X4-MCP:1U |
| +MCP-X4:1M | WH:1(WH) | +AJB-X4-MCP:1M |
| +MCP-X4:1L | SHIELD (PAIR 1) | +AJB-X4-MCP:1L |
| +MCP-X4:2U | BK:2(BK) | +AJB-X4-MCP:2U |
| +MCP-X4:2M | WH:2(WH) | +AJB-X4-MCP:2M |
| +MCP-X4:2L | SHIELD (PAIR 2) | +AJB-X4-MCP:2L |
| +MCP-X4:3U | BK:3(BK) | +AJB-X4-MCP:3U |
| +MCP-X4:3M | WH:3(WH) | +AJB-X4-MCP:3M |
| +MCP-X4:3L | SHIELD (PAIR 3) | +AJB-X4-MCP:3L |
| +MCP-X4:4U | BK:4(BK) | +AJB-X4-MCP:4U |
| +MCP-X4:4M | WH:4(WH) | +AJB-X4-MCP:4M |
| +MCP-X4:4L | SHIELD (PAIR 4) | +AJB-X4-MCP:4L |
| +MCP-X4:5U | BK:5(BK) | +AJB-X4-MCP:5U |
| +MCP-X4:5M | WH:5(WH) | +AJB-X4-MCP:5M |
| +MCP-X4:5L | SHIELD (PAIR 5) | +AJB-X4-MCP:5L |
| +MCP-X4:SHIELD | Shield | +AJB-X4-MCP:SHIELD |

EN 50288-7 XLPE IOS Armoured 5 x 2 x 0,75
Pair 5 spare

| +AJB-X4-MCP |
|---|
| 1U |
| 1M |
| 1L |
| 2U |
| 2M |
| 2L |
| 3U |
| 3M |
| 3L |
| 4U |
| 4M |
| 4L |
| 5U |
| 5M |
| 5L |
| SHIELD |



3-deck through terminal block.

| +MCP-X2 |
|---|
| 1 |
| 2 |
| 3 |
| 4 |
| 5 |
| 6 |
| 7 |

| ACTUATOR-DISCRETE | | |
|---|---|---|
| Lynxeo: 10108697 | | |
| 1.5 mm² (16 AWG) | 25.0 ft | BK |

| | | |
|---|---|---|
| +MCP-X2:1 | 1:BK | +AJB-X2-MCP:1U |
| +MCP-X2:2 | 2:BK | +AJB-X2-MCP:2L |
| +MCP-X2:3 | 3:BK | +AJB-X2-MCP:3L |
| +MCP-X2:4 | 4:BK | +AJB-X2-MCP:4L |
| +MCP-X2:5 | 5:BK | +AJB-X2-MCP:5L |
| +MCP-X2:6 | 6:BK | +AJB-X2-MCP:6L |
| +MCP-X2:7 | 7:BK | +AJB-X2-MCP:7 |

IEC 60502-1 Armoured (SWA) 7 x 1,5
Core 7 spare

| +AJB-X2-MCP | | |
|---|---|---|
| 1U | BU | |
| 2L | BU | |
| 3L | BU | |
| 4L | BU | |
| 5L | BU | |
| 6L | BU | |
| 7 | | |



Use the terminal that is marked blue.
(Upper for T1, lower for 2-6.)

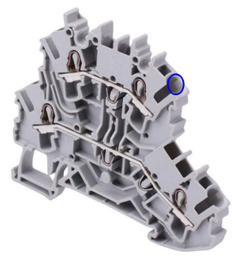
* Terminal 3U means land the wire on the upper deck of the block.
* Note that there are two of each block 3-6.
* (Terminals for field wiring marked BU for clarity.)

| +MCP-X1 |
|---|
| 1 |
| 2 |
| 3 |
| 4 |
| 5 |
| 6 |
| 7 |

| ACTUATOR-DC-POWER | | |
|---|---|---|
| Lynxeo: 10108697 | | |
| 1.5 mm² (16 AWG) | 25.0 ft | BK |
| +MCP-X1:1 | 1:BK | +AJB-X1-MCP:1 |
| +MCP-X1:2 | 2:BK | +AJB-X1-MCP:2L |
| +MCP-X1:3 | 3:BK | +AJB-X1-MCP:3U |
| +MCP-X1:4 | 4:BK | +AJB-X1-MCP:4U |
| +MCP-X1:5 | 5:BK | +AJB-X1-MCP:5U |
| +MCP-X1:6 | 6:BK | +AJB-X1-MCP:6U |
| +MCP-X1:7 | 7:BK | +AJB-X1-MCP:7 |

IEC 60502-1 Armoured (SWA) 7 x 1,5
Core 1 and core 7 spare

| +AJB-X1-MCP | |
|---|---|
| 1 | |
| 2L | BU |
| 3U | BU |
| 4U | BU |
| 5U | BU |
| 6U | BU |
| 7 | |



Use the upper terminal that is marked blue.

* Terminal 3U means land the wire on the upper deck of the block.
* Note that there are two of each block 3-6.
* (Terminals for field wiring marked BU for clarity.)